# Namasivaayam L

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Portfolio GitHub Linkedin

## Professional Summary

Generative AI and Machine Learning Engineer specializing in Multi-modal systems. Orchestrated complex **document verification** workflows using an **autonomous agent** framework, **cutting** processing time by **67%**. Pioneered a Gemini-based solution that boosted document extraction accuracy from **9% to 90%**. Recipient of "**The Rookie**" Award (2025).

## Technical Skills

**AI/ML:** Deep Agents, LangChain, LangGraph, RAG, MCP, PyTorch, TensorFlow, Google ADK

**Certifications: AWS certified Generative AI Developer - Professional**, **GCP Professional Cloud Architect**

**Cloud & DevOps:** GCP, Vertex AI, AWS, Docker, Kubernetes (GKE), Azure Pipelines, GitHub, Rancher, Splunk

**Web & Databases:** FastAPI, Next.js, React, Redis, MySQL, PostgreSQL, MongoDB

**Languages:** Python, JavaScript, C++

## Experience

**Rocket, India (formerly Mr. Cooper)** Jul 2024 - Present

***Software Engineer I***

- **In A Box (Multi-Agent System):** Built and deployed a scalable multi-agent workflow to automate complex multi-document verification across **50+ checkpoints**, achieving a **67% reduction** in manual processing effort.
  - Built a large-scale orchestration graph using **LangGraph** for Loan Modification workflows.
  - Integrated multiple enterprise systems (**FileNet, LSAMS, CMOD)** to enable reliable data aggregation.
  - Implemented cross-verification pipelines across **5+ document types**, combining classification, extraction, and validation within a unified graph.
  - Enabled real-time state tracking and persistence via **MCP tools** interacting with backend services (.NET), ensuring traceability of intermediate and final outputs.
- **PyroAI Docs (GenAI Extraction):** Built a production-grade LLM-based document extraction system, improving accuracy from **9% to 90%** and enabling straight-through processing across **15+ financial document types**, handling **2000+ pages/day**.
  - Migrated legacy Deep Learning-based pipelines to **LLM**-driven architectures and engineered optimized prompt pipelines for diverse document formats and edge cases, significantly improving robustness and maintainability.
  - Reduced monthly cloud infrastructure costs from **$30K+ to < $10K** while decreasing dependency on human-in-the-loop (**HITL**) operations, improving overall system efficiency.
- **AI Guardian (LLM Validation):** Led the development of a scalable **LLM evaluation and validation framework** to benchmark production models against external competitors. Built backend services using **FastAPI, Docker, GKE**.
  - Enabled systematic performance tracking, evaluation, and model comparison across multiple dimensions using **LLM-as-judge**, **Cosine Similarity** and **Levenshtein Distance** metrics.
- **Idea-Code (Autonomous PDLC):** Architected a LangGraph Deepagent pipeline using Azure/GitHub MCP servers to automate the PDLC from concept to GitHub PRs. Orchestrated the autonomous generation of BRDs, architectures, and ADO tasks, streamlining the transition from requirements to production code.

**Rocket, India (formerly Mr. Cooper)** Jan 2024 - Jun 2024

***Software Engineer Intern***

- Delivered a Gemini-based document extraction POC, achieving a **10x accuracy improvement** and establishing the foundation for production adoption.

## Education

**MEPCO Schlenk Engineering College, Tamil Nadu, India** 2020 - 2024

B.Tech in **Artificial Intelligence and Data Science** ***CGPA: 8.49/10***

**Relevant Coursework**: Machine and Deep Learning, Generative AI, Data Science, IoT, Data Analytics, Cloud Computing.

## Awards and Recognition

- **The Rookie (Mr. Cooper, 2025):** Recognized as the top-performing new hire (**Top 1%**) in the PyroAI team for 2024-25.
- **IEEE Conference Publication (2024):** Published "**Pressure, Queue, and Average Speed**-Based **Multi-Agent DQN**" demonstrating superior traffic flow and reduced congestion against fixed-time schedules using **3-tier reward function**.
- **Best Outstanding Student (2024):** Selected as the **sole** recipient out of **50** peers (**Top 2%**) for maintaining an **8.49 CGPA** while **directing** events and operations for the Microsoft Campus Club and Consumer Club.